AC-SINDy: Scalable Sparse Identification of Nonlinear Dynamics via Arithmetic Circuits

Peter Racioppo
Independent Researcher
Los Angeles, CA 90731
pcracioppo@gmail.com

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Abstract

We present AC-SINDy, a compositional extension of the Sparse Identification of Nonlinear Dynamics (SINDy) framework that replaces explicit feature libraries with a structured representation based on arithmetic circuits. Rather than enumerating candidate basis functions, the proposed approach constructs nonlinear features through compositions of linear functions and multiplicative interactions, enabling a compact and scalable parameterization. This formulation trades explicit completeness of the function library for a structured, compositional hypothesis class, in which sparsity is enforced over the computational graph. We further incorporate sequence-based training via Neural Ordinary Differential Equations (Neural ODEs) and explore iterative pruning with a gradient-based importance approximation. Experiments on nonlinear and chaotic systems demonstrate that the method can recover accurate and interpretable models while scaling more favorably than standard SINDy.

1 Introduction

Discovering governing equations of dynamical systems from data is a central problem in scientific machine learning. The Sparse Identification of Nonlinear Dynamics (SINDy) framework addresses this by selecting a sparse combination of candidate basis functions from a predefined library [Brunton et al., 2016], enabling recovery of compact and interpretable models.

However, SINDy suffers from a fundamental scalability limitation: the size of the candidate library grows combinatorially with the state dimension and the desired degree of nonlinearity. Representing higher-order or compositional dynamics therefore requires explicit enumeration of many candidate terms, increasing computational cost and sensitivity to feature design.

In this work, we reformulate SINDy by replacing explicit feature libraries with a compositional, learnable representation. Instead of expressing the dynamics as a linear combination of predefined functions, we construct them implicitly through a structured neural architecture that composes simple functional primitives. This provides a scalable alternative to explicit basis expansion.

In its simplest form, the architecture composes linear transformations and multiplicative interactions, and can be interpreted as a sparse arithmetic circuit. More generally, the framework extends to broader compositional models by incorporating additional nonlinear primitives, allowing richer functional forms to be learned directly from data.

This leads to a shift in the underlying hypothesis class. Standard SINDy represents all candidate functions up to a given complexity, whereas the proposed approach represents a structured subset defined by compositional factorization. This introduces a tradeoff: while not all interactions are

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represented simultaneously, individual interactions can be expressed and recovered under a sparsity assumption.

We further incorporate Neural Ordinary Differential Equations (Neural ODEs) to enable sequence-based training. By optimizing multi-step predictions, the model is encouraged to learn dynamics that are consistent over trajectories, helping to disambiguate competing local solutions.

We also explore sparsification via iterative pruning. Rather than relying on magnitude-based criteria, we evaluate the marginal contribution of parameters to model performance, providing a more direct mechanism for structure learning.

We evaluate the method on a range of nonlinear and chaotic dynamical systems, demonstrating that it can recover accurate and interpretable models. These results suggest that compositional representations provide a promising alternative to explicit feature enumeration for scalable system identification.

Our contributions are as follows:

- We introduce AC-SINDy, a compositional parameterization of SINDy in which candidate functions are represented implicitly via factorized structure rather than explicit enumeration.
- We show that this representation yields polynomial parameter complexity under sparsity assumptions, in contrast to the combinatorial growth of standard SINDy libraries.
- We develop a pruning-based procedure for structure learning in this compositional setting.
- We demonstrate recovery of sparse governing equations in nonlinear and chaotic systems.

2 Related Work

The discovery of governing equations from data has been significantly advanced by the Sparse Identification of Nonlinear Dynamics (SINDy) framework [Brunton et al., 2016], which employs sparse regression over a fixed library of candidate functions to obtain interpretable models. However, this approach is sensitive to noise, depends on the choice of coordinates, and suffers from combinatorial growth in library size.

Several extensions have addressed these limitations. Rudy et al. [2016] improved robustness to noise through regularization and numerical differentiation, while Kaheman et al. [2020] introduced SINDy-PI to handle implicit and rational dynamics. Champion et al. [2019] proposed the SINDy Autoencoder, which learns a latent coordinate system in which the dynamics are sparse, demonstrating that neural networks can aid in discovering parsimonious representations.

A central challenge in SINDy is estimating time derivatives from noisy data. Both et al. [2021] introduced DeePyMoD, which uses automatic differentiation within neural networks to estimate derivatives directly. Forootani et al. [2023] further improve robustness by replacing differential formulations with integral constraints. Related work has also combined SINDy with Neural Ordinary Differential Equations (Neural ODEs), enabling multi-step training and the enforcement of physical structure through backpropagation in time [Lee et al., 2021].

More recent work has begun to relax the fixed-library assumption by learning or adapting candidate functions directly from data. These approaches include parameterized or learned libraries [Yonezawa et al., 2026, Singh and Mukherjee, 2024], as well as architectural extensions such as nested or hierarchical SINDy formulations [Fiorini et al., 2025]. In parallel, advances in neural symbolic regression and compositional architectures, such as Kolmogorov–Arnold Networks, highlight the potential of learned functional representations [Faroughi et al., 2025].

Despite these advances, most existing approaches retain an explicit library of candidate functions or a parametric variant thereof. As a result, they continue to rely on basis expansion, with associated scalability challenges as the state dimension and desired model complexity increase.

In contrast, our approach replaces explicit libraries with a compositional representation based on arithmetic circuits. Rather than enumerating candidate functions, the model constructs features through compositions of linear transformations and multiplicative interactions, defining an implicit hypothesis class. Sparsity is enforced over the structure of this computation, yielding a structured,

factorized representation of the dynamics. This formulation trades explicit completeness of the function library for improved parameter efficiency and scalability.

3 Background: The SINDy Algorithm

Consider a first-order nonlinear dynamical system of the form

$$\dot{x}(t) = f(x(t), u(t), t),$$

where $x(t) \in \mathbb{R}^n$ is the state vector and $u(t)$ denotes control inputs. The Sparse Identification of Nonlinear Dynamics (SINDy) framework seeks to approximate the unknown function f using a sparse linear combination of candidate basis functions.

Specifically, for each component $x_i(t)$, $i \in \{1, \ldots, n\}$, SINDy solves a regression problem of the form

$$\mathcal{L}_i = \|\dot{x}_i(t) - g_i(x(t), t)\|_2^2,$$

where

$$g_i(x(t), t) = \sum_j w_{i,j} f_{i,j}(x(t), t)$$

is a linear combination of predefined basis functions $\{f_{i,j}\}$. The key assumption is that only a small subset of these basis functions is required to accurately describe the system dynamics. The SINDy algorithm is illustrated in Fig. 1.

In practice, SINDy performs least-squares regression combined with iterative thresholding, sequentially removing coefficients with small magnitude (typically based on an L1 criterion). Empirically, this procedure has been observed to outperform standard LASSO-based approaches in identifying sparse and interpretable models.

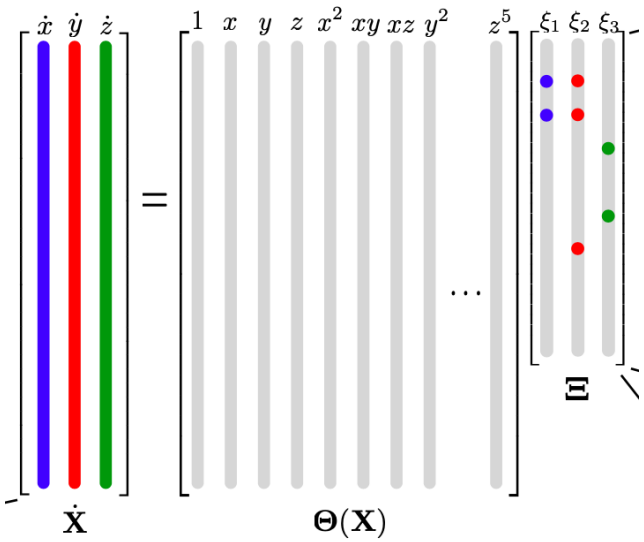

Figure 1: Illustration of SINDy regression, reproduced from the original work.

4 AC-SINDy: Compositional Sparse Identification via Arithmetic Circuits

We propose a compositional extension of the SINDy framework in which the explicit library of candidate functions is replaced by a structured neural architecture. The model constructs nonlinear features through compositions of learned transformations, enabling a compact and scalable representation of dynamical systems.

While the architecture can be interpreted as a sparse arithmetic circuit when restricted to linear and multiplicative operations, this is a special case of a more general compositional framework. Additional nonlinear primitives (e.g., sinusoidal or other parametric functions) can be incorporated, yielding a broader class of neural SINDy models.

4.1 Architecture

Let $x \in \mathbb{R}^n$ denote the system state. At each layer ℓ, we form an augmented input

$$\tilde{x}^{(\ell)} = [x^{(\ell)}; 1] \in \mathbb{R}^{n_\ell+1},$$

where the constant feature enables representation of bias and lower-order terms.

Each layer consists of:

Masked Linear layer.

$$h^{(\ell)} = W^{(\ell)}\tilde{x}^{(\ell)}, \quad W^{(\ell)} \in \mathbb{R}^{d_\ell \times (n_\ell+1)}.$$

Interaction layer. Features are combined via nonlinear interactions. In the simplest case:

$$x_k^{(\ell+1)} = \prod_{j \in \mathcal{G}_k} h_j^{(\ell)},$$

where $\{\mathcal{G}_k\}$ partitions the features. More generally, other nonlinear primitives may be used.

Stacking layers yields a compositional computation graph in which features are built recursively. The compositional structure admits automatic symbolic extraction via propagation of polynomial expressions through the network.

4.2 Function Class

The model represents functions of the form:

$$f(x) = \sum_{k=1}^{m} \psi_k(x),$$

where each ψ_k is constructed by the network. In the multiplicative case:

$$\psi_k(x) = \prod_{j \in S_k} \left(w_{k,j}^T x\right).$$

Unlike standard SINDy, which uses a fixed library $\Theta(x)$, the model defines an implicit hypothesis class through composition and does not enumerate all candidate functions.

4.3 Expressivity and Sparse Compositional Structure

The architecture does not guarantee representation of all polynomial terms simultaneously. For example, a quadratic function

$$f(x) = x^T Q x$$

requires independent coefficients for each pair (i, j), whereas the model represents such functions as

$$f(x) = \sum_{k=1}^{r} (a_k^T x)(b_k^T x),$$

corresponding to a low-rank factorization of Q.

More generally, the model need not span the full function space, but can represent individual interactions or small subsets thereof. This leads to a different assumption: the dynamics admit a sparse compositional representation. Under this assumption, the model can recover the true dynamics by selecting a small number of active terms.

4.4 Interpretation as an Arithmetic Circuit

When restricted to linear and multiplicative operations, the model can be viewed as a sparse arithmetic circuit:

- Linear layers correspond to addition (sum nodes)

- Multiplicative layers correspond to product nodes
- Pruning removes edges in the computational graph

This highlights the role of compositional reuse: intermediate features can be shared across multiple terms.

4.5 Sparsification

Sparsity is enforced by pruning weights in the linear layers. Each parameter corresponds to an edge in the computational graph, and pruning removes edges with minimal impact on performance. This yields a sparse structure in which only a subset of interactions is active.

Unlike standard SINDy, sparsity operates over both feature construction and composition.

4.6 Parameter Efficiency

In standard SINDy, polynomial libraries lead to combinatorial growth. The number of monomials of degree p in n variables is

$$\binom{n+p-1}{p},$$

which counts the number of ways to distribute degree across variables. Including all terms up to order p:

$$\sum_{k=0}^{p} \binom{n+k-1}{k} = \frac{1}{n}(p+1)\binom{n+p}{p+1} \sim \mathcal{O}(pn^p).$$

In contrast, each compositional term requires $\mathcal{O}(pn)$ parameters. If m terms are active, the total parameter count is $\mathcal{O}(mpn)$, yielding $\mathcal{O}(n^2p)$ under $m = \mathcal{O}(n)$.

This improved scaling arises from restricting the hypothesis class to factorized, compositional representations rather than enumerating all monomials. While the model does not represent all terms simultaneously, it is sufficient when the underlying dynamics are sparse.

Figure 2 illustrates this difference in scaling.

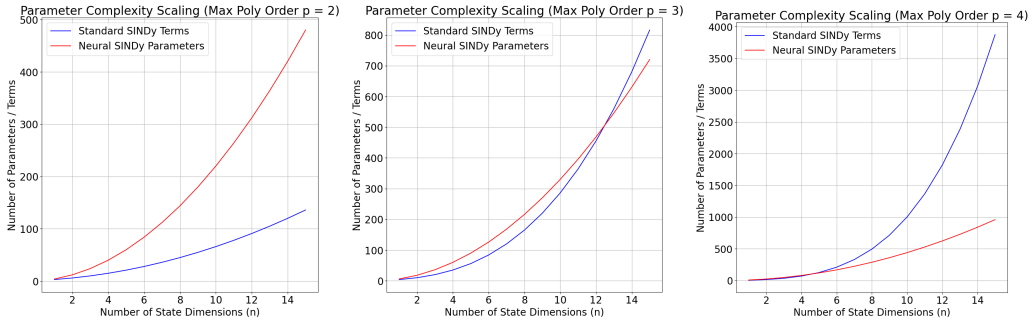

Figure 2: Parameter growth for standard SINDy (explicit enumeration) versus AC-SINDy (compositional representation) for polynomial orders $p = 2, 3, 4$.

5 Importance-Based Pruning in Compositional Models

SINDy enforces sparsity by selecting a small number of active terms from a predefined library. In this setting, each parameter corresponds to a single basis function, and coefficient magnitude provides a reasonable proxy for importance.

In contrast, the proposed model represents functions through a shared computational graph. Parameters correspond to edges that may contribute to many compositional terms, breaking the one-to-one correspondence between parameters and functions. As a result, magnitude-based sparsification is no longer well-aligned with functional importance.

To address this, we consider the marginal contribution of each parameter,

$$\Delta_i = \mathcal{L}(w_{-i}) - \mathcal{L}(w),$$

which measures the effect of removing a parameter on the loss. While this provides a principled notion of importance, computing it exactly is not scalable.

Instead, we approximate Δ_i using a first-order Taylor expansion,

$$\Delta_i \approx \left| w_i \frac{\partial \mathcal{L}}{\partial w_i} \right|,$$

yielding a gradient-based pruning criterion. This captures both parameter magnitude and sensitivity, and scales efficiently in practice. We use this approximation as the primary pruning method. We apply pruning iteratively with finetuning between steps, repeating pruning at convergence.

6 Feature Normalization

A central challenge in both SINDy and the proposed compositional framework is sensitivity to feature scaling. In SINDy, candidate functions such as x, x^2, and higher-order terms can differ by orders of magnitude depending on the region of the state space. In particular, when $|x| \gg 1$, higher-order terms dominate, whereas when $|x| \ll 1$, lower-order terms dominate. As a result, the relative importance of candidate functions depends on the portion of the state space explored during training, and sparsity-inducing procedures may favor certain terms due to scale rather than their true contribution to the dynamics. This mismatch in scaling also distorts the optimization landscape, leading to poorly conditioned gradients and slower convergence. These issues are further exacerbated in compositional models, where intermediate features are constructed through nonlinear combinations and may exhibit highly variable magnitudes.

To address this, we introduce *Feature Normalization*, a simple mechanism that rescales each feature independently using statistics computed over the data. Let $h(x) \in \mathbb{R}^d$ denote the vector of features produced by the model (e.g., the output of the final hidden layer). We define normalized features

$$\tilde{h}_i(x) = \frac{h_i(x)}{\sigma_i + \epsilon},$$

where σ_i is an estimate of the standard deviation of feature h_i over the training trajectory, and ϵ is a small constant for numerical stability. The model output is then given by

$$f(x) = W\tilde{h}(x).$$

This can be interpreted as a scale-invariant parameterization in which coefficients correspond directly to functional importance, independent of feature magnitude. In contrast to standard normalization techniques such as batch normalization, layer normalization, or RMS normalization, Feature Normalization operates independently on each feature and does not mix information across dimensions. This preserves the interpretability of the learned representation, which is essential in the context of system identification.

We compute σ_i from the current batch and update a running estimate using an exponential moving average, but detach σ_i from the computational graph when normalizing features:

$$\tilde{h}_i(x) = \frac{h_i(x)}{\text{stopgrad}(\sigma_i) + \epsilon}.$$

This prevents gradients from flowing through the normalization factors and ensures that parameter updates remain well-behaved.

7 Control with Structured Dynamics

The proposed framework naturally supports model-based control by providing an explicit, interpretable representation of the system dynamics. Given a learned model

$$\dot{x} = g_w(x, u),$$

standard control techniques can be applied, including linearization and trajectory optimization.

A key advantage of AC-SINDy over black-box neural models is the structured and sparse form of g_w. This enables direct reasoning about the dynamics and, in some cases, closed-form controller design. For example, consider a system of the form

$$\dot{x} = p_1(x) + ku,$$

where $p_1(x)$ is approximated by a sparse polynomial. If the desired dynamics are $\dot{x} = p_2(x)$, and the control input is parameterized as a polynomial $u = q_\theta(x)$, then the closed-loop system remains polynomial, and the controller parameters can be chosen to match coefficients:

$$\theta_i \approx \frac{b_i - a_i}{k}.$$

More generally, the compositional structure of the learned model allows for efficient local linearization and application of classical control methods such as LQR. Compared to dense neural representations, the sparse and structured form of AC-SINDy models may lead to more stable and interpretable control policies.

We leave a detailed empirical investigation of control applications to future work.

8 Experimental Results

We implemented the proposed AC-SINDy architecture in PyTorch.[1]

8.1 Experimental Setup

We evaluate the method on low-dimensional dynamical systems, including canonical chaotic and nonlinear systems.

2D nonlinear system.

$$\dot{x}(t) = -0.1x + y, \quad \dot{y}(t) = -2x - 0.1y - 0.5xy - 0.025y^2.$$

Lorenz system.

$$\dot{x} = \sigma(y - x), \quad \dot{y} = x(\rho - z) - y, \quad \dot{z} = xy - \beta z,$$

with $(\sigma, \rho, \beta) = (10, 28, 8/3)$.

Lorenz system with sinusoidal forcing.

$$\dot{x}(t) = \sigma(y - x) + 0.1\sin(x), \quad \dot{y}(t) = x(\rho - z) - y, \quad \dot{z}(t) = xy - \beta z.$$

8.2 Training Dynamics

Figure 3 shows the training loss during iterative pruning. Each pruning step induces a temporary increase in loss, followed by recovery during retraining. For systems with sparse underlying dynamics, pruning initially improves performance by removing redundant parameters, but eventually leads to underfitting. The final model is selected at the minimum validation loss, corresponding to the optimal sparsity–expressivity tradeoff.

8.3 Prediction Quality

Figure 4 shows predictions on the Lorenz system. The model accurately captures both individual state dynamics and the global phase-space structure. As expected for chaotic systems, long-term trajectories diverge from the ground truth.

8.4 Model Recovery

A primary objective of SINDy-based methods is recovery of the underlying governing equations. The proposed approach identifies sparse and interpretable structure, including nonlinear interaction terms.

[1] `https://github.com/PCR-git/AC-SINDy`

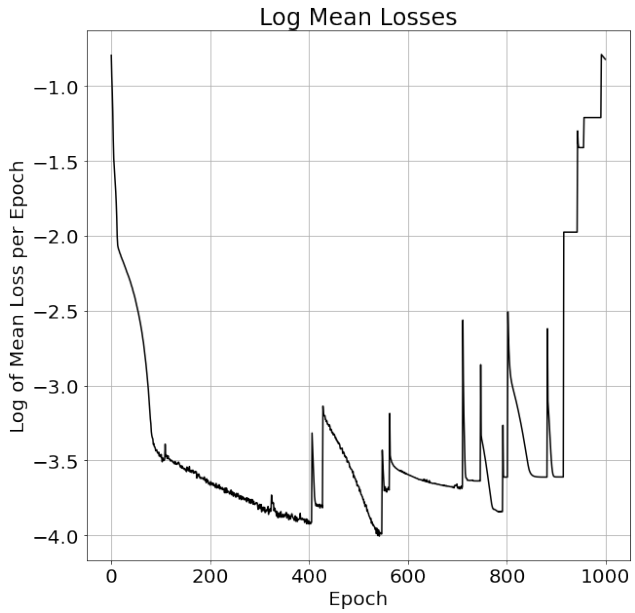

Figure 3: Training loss during iterative pruning. Loss spikes occur when parameters are removed, followed by recovery as the model re-optimizes. Excessive pruning degrades performance.

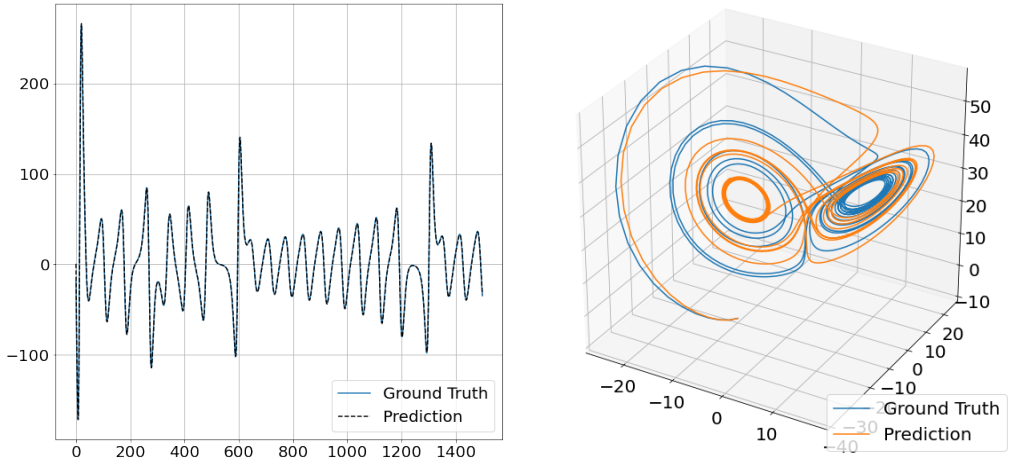

Figure 4: Left: Predicted vs. true trajectory for a single state variable. Right: Phase-space trajectory. The learned dynamics capture the qualitative structure despite long-term divergence.

2D nonlinear system.

$$\text{True dynamics:} \quad \dot{x} = -0.1x + y, \quad \dot{y} = -2x - 0.1y - 0.5xy - 0.025y^2.$$

$$\text{Recovered system:} \quad \dot{x} \approx -0.11x + 1.00y, \quad \dot{y} \approx -1.93x - 0.11y - 0.50xy - 0.029y^2.$$

Lorenz system.

$$\text{True dynamics:} \quad \dot{x} = 10(y - x), \quad \dot{y} = x(28 - z) - y, \quad \dot{z} = xy - \tfrac{8}{3}z.$$

$$\text{Recovered system:} \quad \dot{x} \approx 7.5\,y - 0.6\,x, \quad \dot{y} \approx 24\,x - 0.87\,xz, \quad \dot{z} \approx 0.25\,xy - 0.95\,z.$$

The learned models recover the correct functional structure, including bilinear interactions (xy, xz) and linear terms. Coefficient discrepancies are primarily due to scaling and the compositional parameterization.

Overall, these results demonstrate that the proposed approach can recover accurate dynamical models despite not explicitly enumerating candidate terms, demonstrating that compositional parameterization is sufficient for sparse system identification.

9 Conclusion

We introduced AC-SINDy, a compositional reformulation of SINDy that replaces explicit feature libraries with a structured, factorized representation of nonlinear dynamics. By constructing features through compositions of simple primitives and enforcing sparsity over the resulting computational graph, the method enables interpretable model discovery without explicit basis enumeration. Experiments demonstrate that the approach can recover accurate and compact governing equations.

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